

MineXplore: An Open-Source Reinforcement Learning Exploration Benchmark for GNSS-Denied Underground Environment

Abhishek S*
BuildMachineLabs
abhishekss6363@gmail.com

Badrikanath Praharaj*
BuildMachineLabs
badrikanath.praharaj@gmail.com

Sreeram M.V.*
BuildMachineLabs
mvsreerambl@gmail.com

Abstract—Underground mines present extreme conditions for autonomous robot navigation: GPS is denied, lighting is degraded, and tunnel topology is loop-rich and non-convex. Simulation benchmarks grounded in real production-mine geometry and compatible with GPU-accelerated learning pipelines do not yet exist in the open-source ecosystem. We present MineXplore, an open-source MuJoCo-based navigation benchmark derived from the Leung et al. 2017 Chilean underground copper mine dataset [1]. The environment reconstructs a $104,423 \text{ m}^2$ tunnel network through an six-stage contour-to-MJCF pipeline incorporating octagonal wall cross-sections, LiDAR-sourced jagged wall geometry, three terrain friction zones, a global 5° incline, and periodic spot lighting. Geometric fidelity is validated at an Intersection over Union (IoU) of 0.9538 against the source survey map, and surface texture similarity scores 79.4% across six structural dimensions. A single-agent PPO baseline trained via RLlib across five independent random seeds achieves a best rolling coverage of $88.89\% \pm 1.74\%$ (3 of 5 seeds reaching the 90% coverage target), confirming that MineXplore supports stable and reproducible policy learning under realistic underground sensing and topology.

Index Terms—simulation environment, underground mining robotics, reinforcement learning, MuJoCo, navigation benchmark, exploration

I. INTRODUCTION

Underground mines are among the hardest environments for autonomous ground robots. GPS is denied, lighting is degraded, dust and water reduce sensor returns, and topology is loop-rich and non-convex [3]. The DARPA Subterranean Challenge surfaced both the difficulty of such settings and the community’s best field systems [2]. Yet much of the simulation tooling used for algorithm development remains centred on Gazebo, even as GPU-accelerated simulators have become the default for modern robot learning pipelines [6], [7].

Two gaps motivate this work. First, no publicly released MuJoCo or MJX environment is grounded in a real production mine; existing subterranean simulation assets are predominantly Gazebo-based [4], [8]. Second, rare published real-mine datasets such as the Leung et al. 2017 Chilean underground mine dataset have been used almost exclusively for SLAM evaluation rather than as geometric sources for simulation [1], [3]. MineXplore closes both gaps by compiling the Chilean

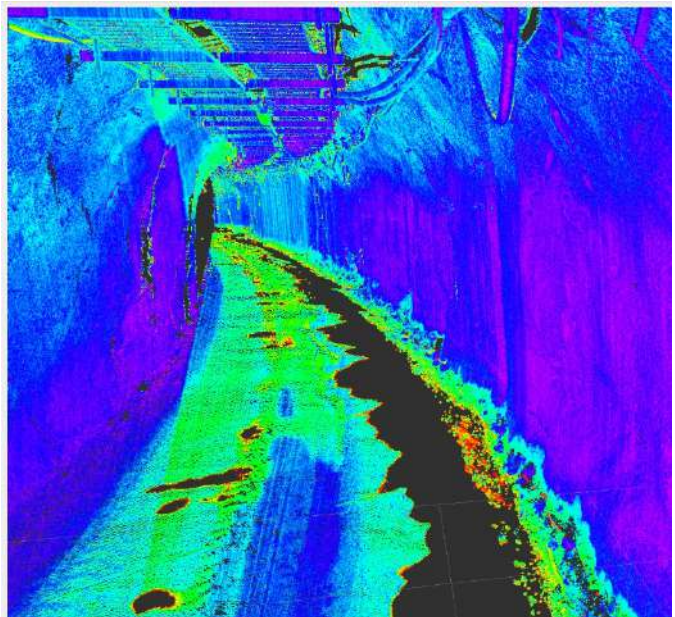


Fig. 1. Point cloud visualisation from the Chilean underground mine survey. The irregular 3D tunnel geometry — narrow corridors, branching junctions, and overhanging rock faces — directly motivates the need for a physics-based benchmark grounded in real mine structure rather than procedural generation.

survey data into a calibrated MuJoCo world directly usable for reinforcement learning under local observations.

Fig. 1 shows a representative point cloud from the Chilean underground mine survey. We present MineXplore, a MuJoCo navigation environment whose geometry is derived from the Leung et al. 2017 dataset [1]. The environment exposes a Gymnasium-style reset/step interface [10] and supports training of reinforcement learning policies under local observations in a topologically complex layout that procedural generators such as BARN do not produce [5]. A full description of the contour-to-MuJoCo XML Format (MJCF) compilation and validation pipeline is provided in Section III.

II. RELATED WORK

A. Underground and Subterranean Robotics

The DARPA Subterranean Challenge (2018–2021) drove recent progress in autonomous systems for GPS-denied un-

*These authors contributed equally to this work.

derground environments, with Team CERBERUS winning the Final Event [2], [4]. Ebadi et al. catalogued open problems in extreme underground SLAM across six SubT teams [3]. The Leung et al. 2017 Chilean underground mine dataset, our geometric source, remains the only publicly released production-mine sensor log suitable for navigation and SLAM research [1].

B. Simulation Environments for Robot Learning

Gazebo-based assets dominate the subterranean simulation niche: the SubT Virtual Testbed ships replica competition courses [4], and BARN provides 300 procedurally generated navigation worlds used in ICRA benchmarks [5]. For large-scale reinforcement learning, Isaac Gym [6] and MuJoCo Playground [7] have become standard.

C. Prior Use of the Chilean Mine Dataset

Prior work has used the Leung et al. dataset exclusively for SLAM benchmarking [3]. To the best of our knowledge, no prior work has compiled its survey geometry into a physics-based simulation environment.

D. Real-Mine-Grounded Simulation

Closest to our work, Gao and Awuah-Offei built a ROS 2 and Gazebo framework for quadruped navigation in 3D maps of real sites including the Edgar Mine teaching facility [8]. MuJoCo, and in particular its GPU-vectorised MJX backend, supports parallel environment stepping at rates orders of magnitude faster than Gazebo, consistent with the throughput expectations of modern RL pipelines [6], [7].

E. Positioning of MineXplore

- **Real production-mine geometry.** Prior assets use procedural generation [5], competition replicas [4], or teaching facilities [8]. MineXplore is the first environment compiled from a real operational mine [1] inside MuJoCo/MJX.
- **Chilean dataset used for simulation.** All prior work uses Leung et al. [1] solely for SLAM benchmarking [3]; MineXplore is the first to convert it into a physics-based RL environment.
- **GPU-scalable over Gazebo.** Gazebo-based frameworks [4], [8] are impractical for deep RL training throughput; MuJoCo/MJX provides the parallel stepping rates required [6], [7].
- **Gymnasium interface over real mine topology.** Unlike BARN [5], Isaac Gym [6], or MuJoCo Playground [7], MineXplore exposes a standard reset/step interface over a 104,423 m² loop-rich, real-mine tunnel network.

The Gymnasium-compatible interface design of MineXplore draws on conventions established by representative simulation environments including BARN [5], Isaac Gym [6], and MuJoCo Playground [7], with the standard reset/step interface provided by the Farama Gymnasium framework [10].

III. ENVIRONMENT DESIGN

Section III walks through the MineXplore compilation pipeline in order: source inputs, 2D geometry extraction, 3D compilation, surface feature and texture mapping, environment realism features, and validation. Stage labels correspond to Fig. 2.

A. Source Data and Scale Calibration (Stage 1)

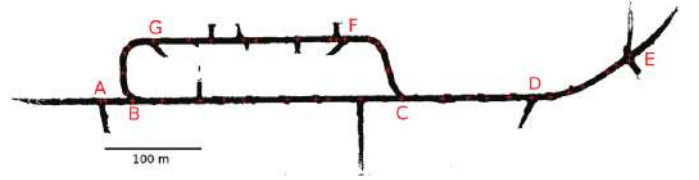


Fig. 3. Published 2D survey floor-plan of the Chilean underground copper mine [1]. Junction density and corridor width in this source map directly determine navigation difficulty in the compiled environment — tighter junctions and narrower passages increase the exploration challenge for the trained agent.

MineXplore uses two complementary inputs [1]. The tunnel geometry is derived from the published 2D survey floor-plan image shown in Fig. 3. The visual surface features—wall roughness, cross-sectional profile, and surface colour—are derived from the accompanying 3D LiDAR point cloud logs. We calibrated the pixel-to-metric scale against the 100 m scale bar on the source image, yielding 1.36 px/m. The nominal tunnel height is set to 3.5 m, consistent with the production tunnel specification in the source dataset. The base floor geometry is a flat horizontal plane with no fine-grained elevation mesh; cross-sectional height variation along the tunnel is not modelled at the mesh level.

B. 2D Geometry Extraction (Stage 2)

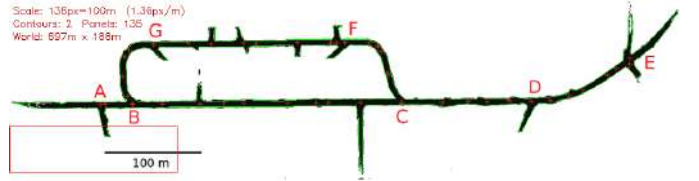


Fig. 4. Detected contours overlaid on the binarised survey map. Outer tunnel boundary in green; interior rock island in red. Accurate contour capture at this stage controls wall placement in the compiled model — the green outer boundary and red rock island correctly enclose the navigable tunnel area, with no false closures or missed passages.

We binarised the survey floor-plan and extracted the tunnel footprint using OpenCV contour detection. The largest external contour defines the outer tunnel boundary; interior contours define rock islands and pillars. Fig. 4 shows the detected contours overlaid on the binarised map. Each contour was converted to a Shapely polygon with holes and simplified using Douglas–Peucker with a tolerance tuned to preserve corners under 1 m. The natural irregularity retained by this tolerance is the primary source of jagged wall geometry in the compiled model. The compiled footprint contains one outer

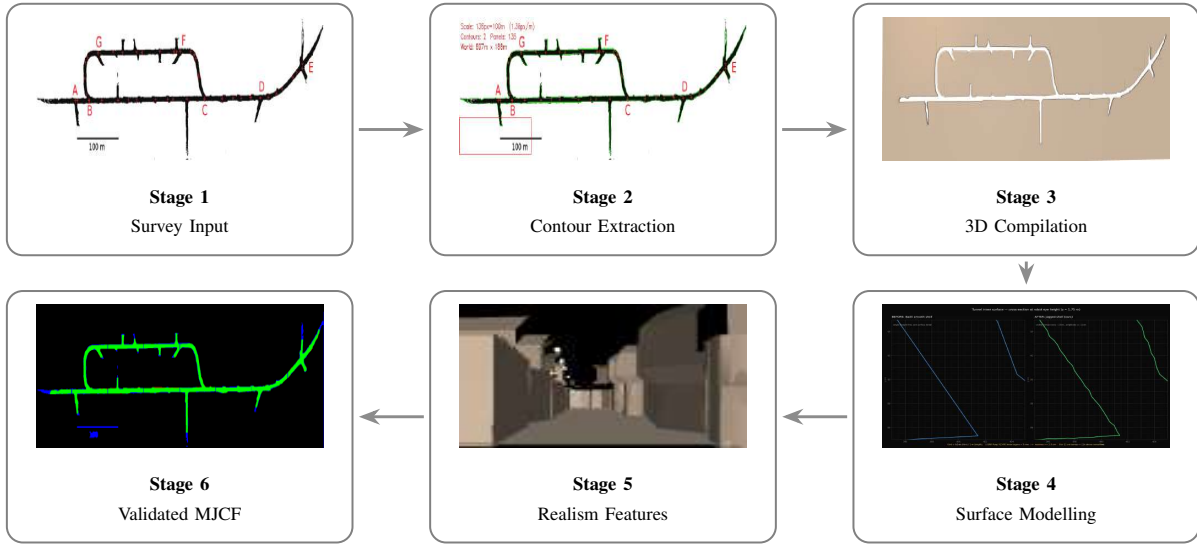


Fig. 2. MineXplore compilation pipeline from real Chilean mine survey data to a validated MJCF environment. Stage 1: 2D survey floor-plan and LiDAR point cloud as source inputs. Stage 2: binarisation and OpenCV contour detection of the tunnel boundary and rock island. Stage 3: polygon triangulation, 3.5 m extrusion, and V-HACD convex decomposition into 1,186 collision geoms. Stage 4: two-layer visual wall system comprising an 8-sided polygonal base layer and a LiDAR-sourced jagged overlay. Stage 5: three realism features applied to the MJCF worldbody. Stage 6: orthographic top-down IoU validation against the source survey map ($\text{IoU} = 0.9538$). Each stage produces a validated intermediate output that feeds the next.

boundary and one dominant interior rock island of 14,577 m², yielding a navigable area of 104,423 m² within a world extent of approximately 697 m $\times$ 188 m.

C. 3D Compilation and Physics Setup (Stage 3)

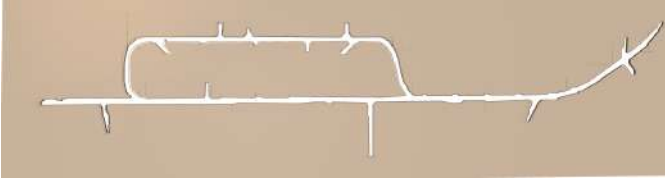


Fig. 5. Compiled MineXplore environment in the MuJoCo interactive viewer. Confirms that the geometry pipeline introduces no systematic distortions to tunnel structure or junction connectivity.

We triangulated the 2D polygon set and extruded each face to 3.5 m. MuJoCo’s collision engine requires convex geoms; we decomposed the extruded mesh using V-HACD with a resolution of 1,000,000, a concavity threshold of 0.5, and a maximum of 32 convex hulls per rock tile. Applied across 70 rock geometry tiles, this produces 1,186 convex collision geoms in the final MJCF model. The compiled world is loaded and centred on the main tunnel network. Ceiling and floor planes close the tunnel volume. Physics parameters set baseline wall and floor friction and include an occupancy octomap for collision-free spawn sampling. Compilation and scene load complete in under ten seconds on an AMD Ryzen 5 5600H under Ubuntu Linux. Fig. 5 shows the compiled MuJoCo world.

D. Surface Wall Geometry and Visual Detail (Stage 4)

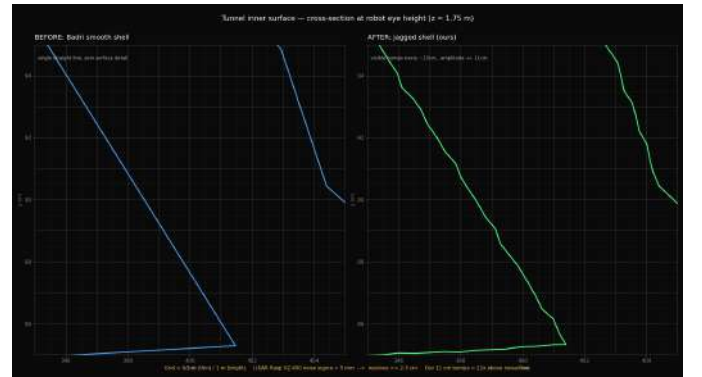


Fig. 6. Cross-sectional wall profile comparison: smooth flat-extruded geometry (left) versus the two-layer surface model (right). The two-layer model adds wall irregularity absent in the flat-extruded baseline; higher wall surface complexity increases LiDAR return variation, making sensor-based navigation harder and more representative of real mine conditions.

The base layer is an 8-sided polygonal mesh that approximates the characteristic bore cross-section of the production tunnel. The overlay layer is a jagged mesh derived from the Leung et al. 3D LiDAR point cloud that captures the centimetre-scale rock surface irregularity of the real tunnel walls. Both mesh geoms are configured as purely visual and do not participate in collision detection; physics contact is handled exclusively by the V-HACD convex hull decomposition. This two-layer design separates physical accuracy from visual realism and avoids the computational cost of simulating fine-surface contact. A uniform rock material is applied across all 1,186 rock meshes and both wall layers. Fig. 6 illustrates the

difference between the original flat-extruded geometry and the two-layer surface model.

E. Environment Realism Features (Stage 5)

MineXplore incorporates three additional features that replicate the physical conditions of the Chilean copper mine.

Terrain incline. The entire MJCF worldbody is rotated 5° (0.0872665 rad) around the Y -axis, tilting walls, floor, rocks, and lights uniformly as a rigid body. This produces an effective elevation gain of 5.2 m across the tunnel’s entrance region, consistent with the terrain gradient in the source dataset, without modifying the V-HACD collision decomposition.

Terrain friction zones. Three spatially fixed floor box geoms partition the tunnel along its east–west axis to replicate mine water seepage: a dry western zone (friction 1.00), a muddy central zone (friction 0.55), and a waterlogged eastern zone (friction 0.25), each covering $119\text{ m} \times 102\text{ m}$. The contact friction compounds with the actuation-level scaling in Section IV to jointly model traction degradation across ground conditions.

Periodic lighting. Thirty-three MuJoCo spot lights are placed along the tunnel centreline at 20 m intervals, 3.0 m above the floor, directed slightly downward with a 45° cone cutoff.

F. Geometry & Surface Validation (Stage 6)

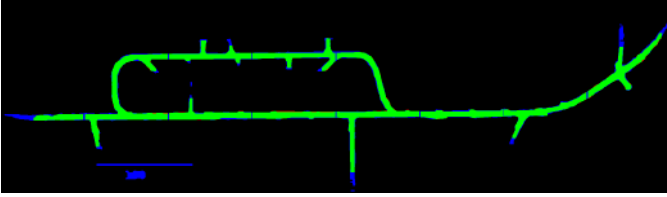


Fig. 7. Pixel-wise geometric fidelity overlay. Green pixels indicate agreement; blue pixels indicate deviation. Green pixels (95.3%) confirm geometric agreement with the source map; blue deviation pixels are concentrated at junctions and corridor edges as expected from the Douglas–Peucker simplification, and are not distributed randomly across the tunnel interior.

We validated the compiled model on two axes. For navigability, we cast rays through the free volume and confirmed that every pixel marked navigable in the source floor-plan maps to an unobstructed column in the MJCF model. For geometric consistency, we rendered an orthographic top-down projection of the compiled MJCF model, binarised it, and compared it pixel-wise against the original survey floor-plan. Fig. 7 shows the resulting overlay: green pixels mark agreement; blue pixels mark deviation, concentrated at junctions and corridor edges where the Douglas–Peucker simplification rounds tight corners. The strict Intersection over Union (IoU) after noise cleanup is 0.9538, confirming that the compiled model reproduces 95.3% of the source survey footprint at pixel level. This overlay is produced directly from the compiled MJCF, not from any intermediate representation.

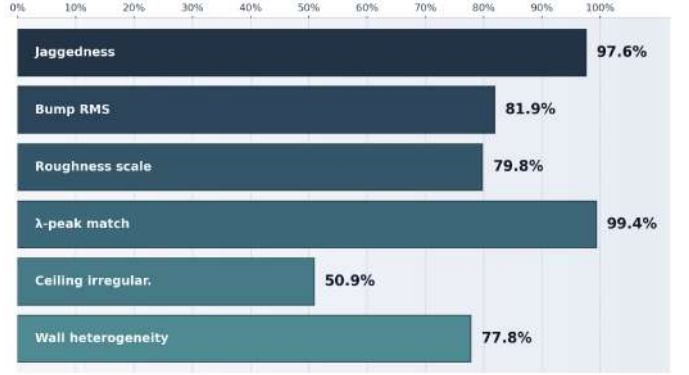


Fig. 8. Per-axis surface-texture similarity comparison between the compiled MineXplore model and the real Chilean mine LiDAR point cloud. All six dimensions score above 50%; dominant wavelength (99.4%) and wall jaggedness (97.6%) are near-perfect matches, confirming that the spatial frequency structure of the real mine walls is faithfully reproduced at the scales most relevant for LiDAR-based navigation.

We assessed surface texture fidelity by comparing the compiled MineXplore wall geometry against the real 3D LiDAR point cloud by fitting SVD-based principal planes over a voxel-scale sweep $w \in \{0.10, \dots, 2.00\}$ m, computing ISO 4287-style residuals (R_a , R_q , R_z , p_{95}), and extracting a 1D along-wall wavelength spectrum via arc-length-resampled FFT across 5 cm horizontal slabs. Six structural dimensions were scored on a per-axis similarity metric (Fig. 8): wall jaggedness (97.6%), bump R_q (81.9%), roughness scale integral (79.8%), dominant wavelength λ -peak (99.4%), ceiling irregularity (50.9%), and wall heterogeneity (77.8%), yielding an overall geometric-mean score of 79.4%. The simulated surface undershoots real roughness magnitude by $\approx 2.5\times$ at the half-metre scale—an intentional amplitude clip to prevent narrow passages from pinching shut—while faithfully replicating the dominant 5–20 cm spatial-frequency band of the real mine walls.

IV. BENCHMARK INTERFACE

A. Observation Space

Each agent receives a 19-dimensional observation vector at every control step. The first 16 elements are LiDAR range readings from a simulated 2D scanner with 16 uniformly spaced beams covering the full 360° field of view ($[0, 2\pi)$). Raw ranges are clipped to $[0.12\text{ m}, 10.0\text{ m}]$, then normalised to $[0, 1]$. Gaussian noise with $\sigma = 0.01\text{ m}$ is applied to each beam before normalisation. The remaining 3 elements are: the previous normalised linear command, a fixed scalar of 0.0, and the previous normalised angular command, providing the agent with one step of action history.

B. Action Space

The action space is a continuous 2D box $\mathcal{A} = [-1, 1]^2$. The policy output (a_0, a_1) is mapped to differential-drive commands as:

$$v = a_0 \cdot 1.5\text{ m/s}, \quad \omega = a_1 \cdot 1.5\text{ rad/s} \quad (1)$$

Commands are then scaled and perturbed by terrain type before execution: dry terrain applies scale 1.0 with zero noise; muddy terrain applies scale 0.7 with additive noise $\sigma = 0.04$ m/s; waterlogged terrain applies scale 0.4 with noise $\sigma = 0.10$ m/s. This actuation-level scaling compounds with the MuJoCo contact friction values described to model the full traction degradation across terrain types.

C. Episode Structure

MineXplore is a pure exploration environment. There is no goal position: `terminated` is always `False` and episodes end only by truncation. The maximum episode horizon is 6,000 control steps at a timestep of 0.1 s, giving a nominal maximum duration of 600 s. An episode is also truncated early if the agent’s speed remains below 0.01 m/s for 100 consecutive steps, preventing degenerate stationary policies. At the start of each episode, the agent is spawned at a fixed anchor near the mine entrance; if that cell fails clearance checks, the agent falls back to a random valid point in the entrance region with uniform yaw sampling over $[-\pi, \pi)$.

D. Reward Structure

The per-step reward r is:

$$r = \Delta c \cdot g - \mathbb{1}[\Delta c = 0] \cdot 0.002 + M - \mathbb{1}[\text{collision}] \cdot 15.0 \quad (2)$$

The reward is structured around a dense exploration signal combined with hard safety and efficiency constraints. Without a per-step coverage term, initial experiments showed coverage saturating near 40%: the agent circled already-mapped corridors since no per-step incentive existed to push into undiscovered regions, producing a flat reward curve beyond early training. The $\Delta c \cdot g$ term resolves this by rewarding only *newly* discovered cells at each step — using a delta rather than cumulative coverage ensures the agent receives signal exclusively for active exploration and not for revisiting mapped space. The per-cell gain $g = 10.0/C_{\text{total}}$ normalises reward magnitude by total tunnel area, making it resolution- and map-size-invariant. The step penalty $-\mathbb{1}[\Delta c = 0] \cdot 0.002$ discourages stationary behaviour. Milestone bonuses M at 25% (+15), 50% (+30), and 75% (+45) provide curriculum-style waypoints that prevent the policy stalling at local coverage maxima during early training. The collision penalty of -15.0 is set at $7,500\times$ the step penalty, creating an unambiguous priority ordering: wall avoidance dominates over time efficiency, which dominates over standing still. We use RL solely to validate that MineXplore supports stable exploration under local observations; it is not a contribution of this paper.

E. Reproducibility

In the experiments reported here we use a single-agent configuration. All experiments use MuJoCo 3.2.6 [12]. Training is run across five independent random seeds (42, 142, 242, 342, 442); all metrics are reported as mean $\pm$ standard deviation across seeds. The compiled MJCF model is kept under version control. Code and environment pipeline will be made open-source upon publication.

V. RESULTS

A. Reinforcement Learning Baseline

We use a single-agent PPO policy, trained via RLlib, to validate that MineXplore supports stable exploration learning under local observations. RL is not a contribution of this paper; the baseline exists solely to confirm that the environment is navigable and that reward signals are well-formed. Training was run across five independent random seeds with the reward structure defined in Section IV-D. Per-seed episode counts varied by stopping condition: three seeds (42, 342, 442) terminated upon reaching the coverage target at episodes 257, 432, and 282 respectively; two seeds (142, 242) triggered the early-stop criterion after coverage regression at episodes 962 and 416. All metrics are reported as mean $\pm$ standard deviation across all five seeds over the final 100 episodes of each run.

TABLE I
PER-SEED TRAINING OUTCOMES. STOP: TC = TARGET COVERAGE REACHED; ED = EARLY STOP DUE TO COVERAGE DROP. BESTROLL = BEST ROLLING COVERAGE.

Seed	Stop	Ep.	Final Cov. (%)	BestRoll (%)
42	TC	257	100.00	89.79
142	ED	962	70.05	85.43
242	ED	416	70.77	89.39
342	TC	432	89.83	89.95
442	TC	282	92.00	89.88
Mean	—	—	84.53 $\pm$ 12.02	88.89 $\pm$ 1.74

Table I summarises the per-seed outcomes. Three seeds (42, 342, 442) reached the coverage target and stopped cleanly. Two seeds (142, 242) triggered early-stop due to coverage regression. Despite early stopping, all five seeds achieved a best rolling coverage between 85.43% and 89.95%, giving an aggregate best rolling coverage of $88.89\% \pm 1.74\%$.

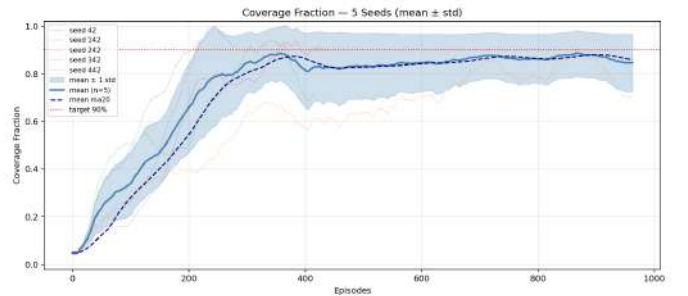


Fig. 9. Coverage fraction over training (mean $\pm$ 1 SD across five seeds; dotted line: 90% target). Mean rises monotonically from $\sim 5\%$ to a best rolling average of $88.89\% \pm 1.74\%$; higher is better.

Fig. 9 shows the mean coverage fraction across all five seeds. The mean rises monotonically from $\sim 5\%$ at episode 1 and approaches the 90% target threshold, with the shaded band confirming consistent behaviour across seeds. The two early-stopped seeds account for the slight downward drift in the mean after episode 400, which is itself informative: even

without reaching the target, those seeds sustain above-85% best rolling coverage.

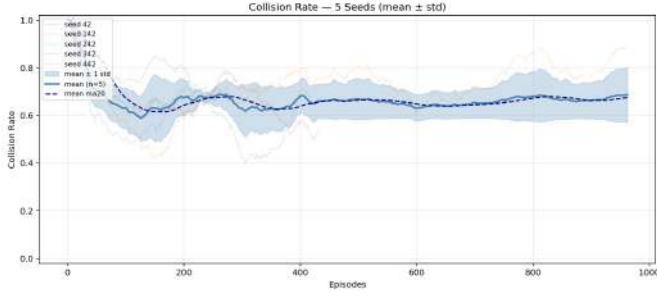


Fig. 10. Per-episode collision rate over training (mean $\pm$ 1 SD across five seeds). Mean falls from ~ 1.0 to 0.65 ± 0.13 in the final episodes; lower is better.

Fig. 10 shows the per-episode collision rate. The mean rate begins near 1.0 and falls to 0.65 ± 0.13 in the final episodes. The persistent collision rate reflects the difficulty of navigating V-HACD collision geometry under purely local LiDAR observations; the declining trend across all five seeds confirms that the environment produces a consistent learning signal for wall avoidance.

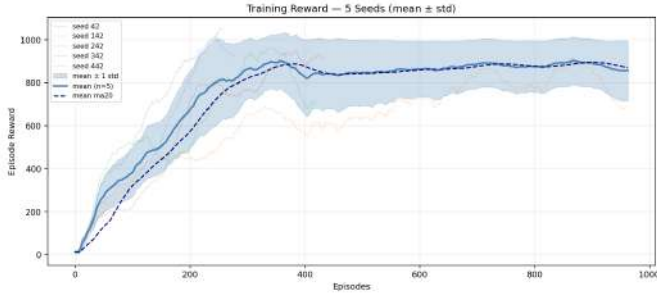


Fig. 11. Cumulative episode reward over training (mean $\pm$ 1 SD across five seeds). Rises monotonically to 860 ± 150 ; higher is better; alignment with coverage confirms no degenerate exploitation across any seed.

Fig. 11 shows episode reward, which rises monotonically and stabilises around 860 ± 150 . The alignment with coverage growth confirms that reward growth is coupled to exploration breadth as designed, with no degenerate reward exploitation visible across any of the five seeds.

VI. DISCUSSION

MineXplore is a single-mine MuJoCo/MJX navigation environment grounded in a real production tunnel, targeted at researchers who need a non-procedural, topology-rich underground test case for navigation and exploration algorithm development. It is not a physics-calibrated digital twin: wall friction, surface roughness, and contact stiffness are approximations derived from the source data, not direct measurements. The collision model is a V-HACD convex decomposition, which may introduce localised hull artefacts in the narrowest tunnel passages. The elevation model is a single global rotation

approximating the tunnel gradient; fine-grained topographic variation and cross-sectional height changes along the full tunnel length are not modelled and are deferred to future work using the Leung et al. 3D LiDAR elevation logs. The five-seed evaluation reveals meaningful variance in coverage outcomes ($\pm 12.02\%$ final, $\pm 1.74\%$ best rolling), with two seeds triggering early stops due to coverage regression — a property that distinguishes strong from weak exploration initialisations and is desirable in a benchmark intended to stress-test exploration algorithms.

MineXplore is the right tool when a large-scale, real-mine-derived obstacle topology inside a MuJoCo or MJX training pipeline is needed, alongside a Gymnasium-compatible interface for single- or multi-agent navigation under local observations, or matched source data for future sim-to-real work grounded in the Leung et al. Chilean mine dataset [1]. Future work includes full elevation-aware geometry from the Leung et al. 3D LiDAR logs, LiDAR-replay validation against the original dataset sweeps, procedural cross-section perturbation for environment diversity, broader multi-agent and cross-algorithm baselines & point cloud data verification.

VII. CONCLUSION

We presented MineXplore, an open-source MuJoCo navigation environment derived from the Leung et al. 2017 Chilean underground copper mine survey. We described the six-stage contour-to-MJCF compilation pipeline (Fig. 2), validated the compiled geometry against the source survey via a top-down overlay (Fig. 7), and confirmed navigability using a single-agent PPO baseline that achieves a best rolling coverage of $88.89\% \pm 1.74\%$ with a mean final coverage of $84.53\% \pm 12.02\%$ across five independent seeds. MineXplore closes a specific gap in the robot-learning tooling landscape: a real-mine-grounded, Gymnasium-compatible navigation environment in the MuJoCo ecosystem, augmented with LiDAR-sourced wall geometry, terrain heterogeneity, a global incline, and periodic spot lighting. Future work includes full elevation-aware geometry from the Leung et al. 3D LiDAR logs, LiDAR-replay validation against the original dataset sweeps, procedural cross-section perturbation for environment diversity, and broader multi-agent and cross-algorithm baselines.

ACKNOWLEDGMENT

The authors thank BuildMachineLabs for providing the opportunity, research direction, and resources that supported this work. The authors also thank Leung et al. for releasing the Chilean underground mine dataset [1], which forms the geometric basis of MineXplore.

REFERENCES

- [1] K. Y. K. Leung, D. Luhr, H. Houshiar, F. Inostroza, D. Borrmann, M. Adams, A. Nüchter, and J. Ruiz-del-Solar, “Chilean underground mine dataset,” *The International Journal of Robotics Research*, vol. 36, no. 1, pp. 16–23, 2017. DOI: 10.1177/0278364916679497.
- [2] M. Tranzatto, T. Miki, M. Dharmadhikari, L. Bernreiter, M. Kulkarni, F. Mascari, O. Andersson, S. Khattak, M. Hutter, R. Siegwart, and K. Alexis, “CERBERUS in the DARPA Subterranean Challenge,” *Science Robotics*, vol. 7, no. 66, p. eabp9742, 2022.

- [3] K. Ebadi, L. Bernreiter, H. Biggie, G. Catt, Y. Chang *et al.*, “Present and future of SLAM in extreme environments: The DARPA SubT Challenge,” *IEEE Transactions on Robotics*, vol. 40, pp. 936–959, 2024.
- [4] Open Robotics, “DARPA SubT Virtual Testbed,” 2021. [Online]. Available: <https://github.com/osrf/subt>
- [5] D. Perille, A. Truong, X. Xiao, and P. Stone, “Benchmarking metric ground navigation,” in *Proc. IEEE Int. Symp. Safety, Security, and Rescue Robotics (SSRR)*, 2020.
- [6] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, “Isaac Gym: High performance GPU based physics simulation for robot learning,” in *NeurIPS Datasets and Benchmarks Track*, 2021.
- [7] K. Zakka, B. Tabanpour, Q. Liao, M. Haiderbhai, S. Holt, J. Y. Luo, A. Allshire, E. Frey, K. Sreenath, L. A. Kahrs, C. Sferrazza, Y. Tassa, and P. Abbeel, “MuJoCo Playground,” *arXiv preprint arXiv:2502.08844*, 2025.
- [8] Y. Gao and K. Awuah-Offei, “Navigation in underground mine environments: A simulation framework for quadruped robots,” in *Proc. IEEE Int. Conf. Automation Science and Engineering (CASE)*, 2025, pp. 1464–1469.
- [9] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal Policy Optimization Algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [10] M. Towers, A. Kwiatkowski, J. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão, A. Kallinteris, M. Krimmel, A. KG, R. Perez-Vicente, A. Pierré, S. Schulhoff, J. J. Tai, H. Tan, and O. G. Younis, “Gymnasium: A Standard Interface for Reinforcement Learning Environments,” *arXiv preprint arXiv:2407.17032*, 2024.
- [11] E. Liang, R. Liaw, R. Nishihara, P. Moritz, R. Fox, K. Goldberg, J. Gonzalez, M. Jordan, and I. Stoica, “RLlib: Abstractions for Distributed Reinforcement Learning,” in *Proc. 35th Int. Conf. Machine Learning (ICML)*, 2018, pp. 3053–3062.
- [12] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
- [13] B. Yamauchi, “A frontier-based approach for autonomous exploration,” in *Proc. IEEE Int. Symp. Computational Intelligence in Robotics and Automation (CIRA)*, 1997, pp. 146–151.
- [14] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations: Theory and application to reward shaping,” in *Proc. 16th Int. Conf. Machine Learning (ICML)*, 1999, pp. 278–287.
- [15] M. G. Bellemare, S. Srinivasan, G. Ostrovski, T. Schaul, D. Saxton, and R. Munos, “Unifying count-based exploration and intrinsic motivation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 29, 2016.